Did the Wolfe Trahan relationships predict company results?

A firm-level test of eight 2011 claims on US non-financial companies, 1994–2026

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Run 24 October 2026. Decision page: `docs/decisions/2026-10-24-wt-relationship-tests.md`

The answer

1. **Mostly no.** We ran 87 tests on 17,767 companies. Only 3 passed every rule we set in advance. Only 1 survives the strictest correction for running 87 tests.
2. **One small effect is real.** When factory input prices jump, the costs of highly leveraged industrial companies rise. The rise is about 0.2% of sales within two quarters, for each standard deviation of the jump. It held before and after 2011.
3. **The sales claim was backwards.** The notes say input-price jumps come before slower sales. In fact, sales sped up after a jump ($t = +5.2$ after 2011). The jumps came at the start of recoveries.
4. **Two directions hold, but they do not improve forecasts.** High inflation went with weaker consumer EBITDA and margins. Rising oil went with higher energy EBITDA and higher consumer costs. Both held the claimed way in both periods. Neither added enough to a forecast to pass.
5. **Two of the model's own macro inputs failed after 2011.** Bank lending standards and the factory survey did not predict company EBITDA declines after 2011. The Baa spread did a little.
6. **Company sensitivities persist for some signals.** Companies that reacted more to input prices or to the Korean stock market in the past reacted more again. This supports a sensitivity per issuer, not one per sector.

What to do. Add none of these series to the model's forecast inputs. Use the confirmed directions (H2, H3) only to set signs in the stress scenarios. Open a decision page to review the macro inputs of the downside model (methodology 5.2).

The question

Each claim says: when a macro series moves, the companies in a sector later report different numbers. We asked whether that happened in history, company by company. A claim is useful only if it predicts *company* results. Those results drive leverage, interest cover, and default risk.

1 What we tested

The 2011 notes chart pairs of series. We kept the eight claims that map to company results (decision page, "The eight claims to test"). The sign and the lead of each claim come from the note. We did not search over leads.

Claim	Signal	Claimed effect on companies
H1	Factory prices paid (Philadelphia Fed)	Costs rise within 2 quarters; sales slow within 4 quarters. Industrials, materials, consumer
H1c	H1, 6-month jump instead of level	Same as H1. A cross-check: the first run used this signal
H2	CPI inflation, year on year	Consumer EBITDA and margin fall; energy and materials EBITDA rise
H3	Oil price, year on year	Energy EBITDA rises; costs of transport and consumer companies rise
H4	Housing starts, year on year	Homebuilders' and building-product makers' EBITDA and sales rise
H5, H5k	Copper; Korean stock market (KOSPI)	Sales of materials and industrial companies speed up within 2 quarters
H6	Factory new orders (Philadelphia Fed)	Sales and EBITDA of industrials, materials, autos rise
H7, H7n	Real yield curve (10-year minus CPI); nominal curve (10-year minus 3-month)	A steeper curve means fewer large EBITDA declines a year ahead
H8a–c	Baa spread change; bank lending standards; factory activity	The model's current inputs: more (fewer) large EBITDA declines a year ahead

2 Data and method

1. **Companies.** Compustat quarterly, all US non-financial companies (every GICS sector except financials and real estate), 1992–2026. A company enters a quarter when its sales over the last four quarters exceed $40m. That gives 17,767 companies and 446,255 company-quarters. Each company counts once. So the test reflects the typical company, not the largest ones.
2. **Outcomes, per company.** All measures use four-quarter totals, which removes seasonal swings. All measures cover the quarters *after* the signal quarter: cost ratio (cost of goods sold over sales), change over 2 quarters; EBITDA margin, change over 4 quarters; EBITDA change over 4 quarters, as a share of sales; sales growth, next 4 (or 2) quarters minus the previous 4 (or 2); leverage (debt over EBITDA); and a large-decline event, EBITDA down by more than 20% in 4 quarters. Extreme values are trimmed at 1% and 99% each quarter.
3. **Signals, as known then.** Each macro value enters only when published. CPI and housing starts enter one month late. Each signal is a z-score against its own history up to that quarter.
4. **High-leverage companies.** Debt above 4 times EBITDA. These are the companies closest to high-yield issuers.
5. **Three tests.**
   - **A: the average company in the group.** The equal-weighted mean outcome each quarter. Run on all companies, and on high-leverage companies only.
   - **B: the group against a comparison group, in the same quarter.** For example, consumer companies minus energy and materials companies. The common economic swing drops out. This equals a company panel with a fixed effect for each quarter.
   - **C: each company's own sensitivity.** Each quarter, estimate how much each company reacted to the signal in its own past. Then ask whether those that reacted more before react more again (Fama–MacBeth). A slope of 1 means full carry-over. A slope of 0 means none.

6. **Statistics for A and B.** The study runs a regression of the outcome on the signal. The controls are factory activity, the Baa spread change, bank lending standards, and the outcome's own recent past. Newey–West errors correct for the overlapping horizons. The study runs the regression on 1994–2011 (the years the notes could see) and on 2012–2026 (the true test). A forecast race from 2012 compares a model with the signal against the model without it (Clark–West). The study also counts the signal episodes and checks the four cycle windows (2008–09, 2015–16, 2020, 2022).

7. **Pass rule (set in advance).** Right sign after 2011. A forecast gain with $p < 0.05/8$. The claimed sign in at least 3 of the 4 cycles, and in at least 2 of the 3 after 2011.

3 Results

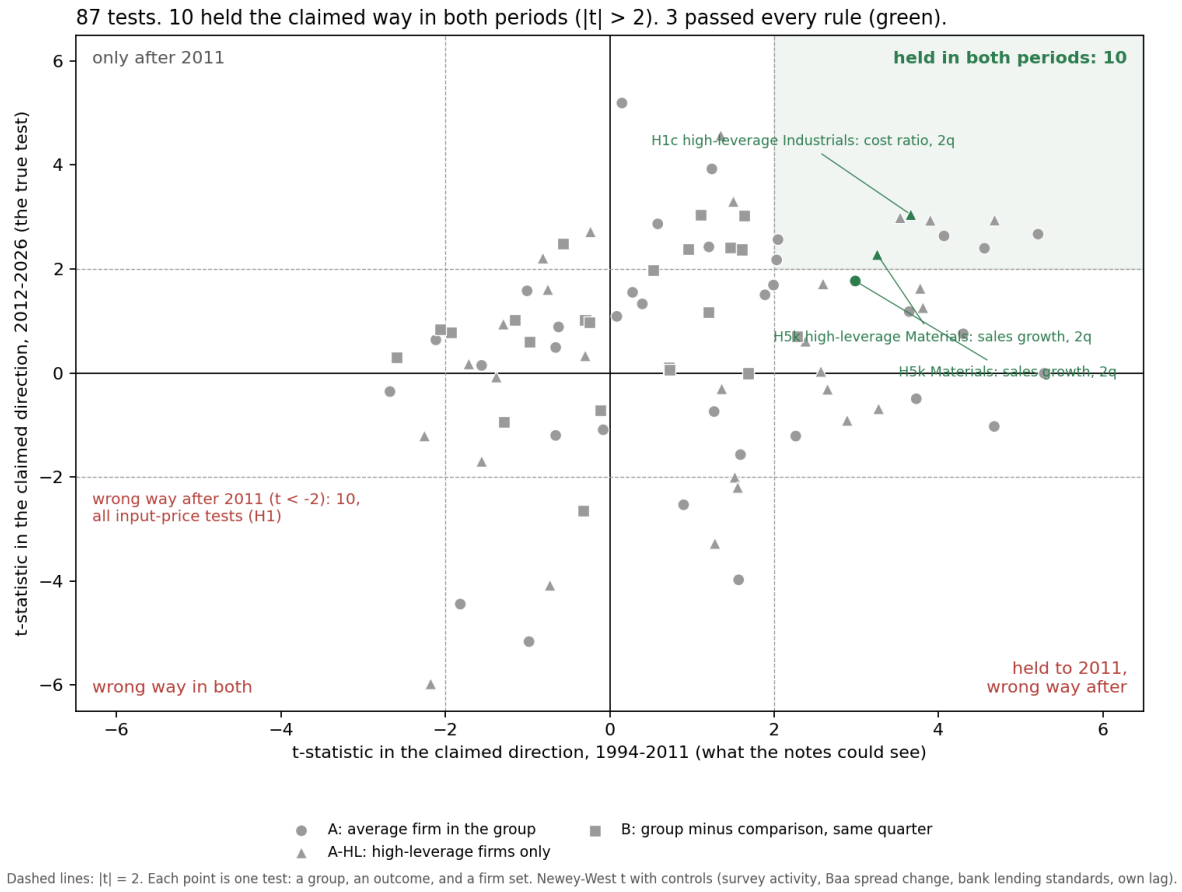


Figure 1: Each point is one test. The horizontal axis shows the evidence in the years the notes could see. The vertical axis shows the evidence after 2011. A useful claim sits top right. Ten tests sit there. Ten tests went clearly the wrong way after 2011, and all ten are input-price tests.

Claim by claim

Claim	Verdict	Evidence
H1	Failed	Costs of the average industrial and consumer company rose after high prices paid to 2011 (t +4.7, +4.3). After 2011 the effect is near zero. The sales claim fails. 0 of 18 tests pass.
H1c	One pass	High-leverage industrials: costs up 0.22 points of sales per 1 sd jump, within 2 quarters (t +3.7 to 2011, +3.1 after; forecast gain $p = 0.0001$; 3 of 4 cycles). Sales sped up instead of slowing (t +3.3 to +6.0 against the claim, after 2011).
H2	Direction holds	Consumer EBITDA fell with high inflation (t −5.2 to 2011, −2.7 after). Margin too (−4.6, −2.4). 3 of 4 cycles. No forecast gain ($p = 0.12$, 0.09). The commodity side and the consumer-minus-commodity test fail.
H3	Direction holds	Energy EBITDA rose with oil, strongest for high-leverage energy companies after 2011 (t +4.6; $p = 0.014$). Consumer costs rose (t +4.1, +2.6; $p = 0.036$). Transport fails.
H4	Failed	Housing-linked EBITDA followed housing starts only after 2011 (t +0.2 to 2011, +5.2 after). About 55 companies a quarter.
H5	Failed	Copper did not predict sales.
H5k	Passes the rule	Materials sales sped up 2.8 points per 1 sd of KOSPI growth, within 2 quarters (t +3.0 to 2011, +1.8 after; $p = 0.006$; high leverage $p = 0.0006$). Just misses the strictest correction.
H6	Failed	New orders predicted neither sales nor EBITDA, for any group.
H7	Failed	A steep real curve went with fewer declines after 2011 (t −3.9) but not before (−1.2). No forecast gain. The nominal curve fails too.
H8a	Weak	A wider Baa spread went with more declines. For high-leverage companies t is +3.5 to 2011 and +3.0 after ($p = 0.04$). For all companies, 4 of 4 cycles ($p = 0.016$). Closest of the model's inputs.
H8b	Failed after 2011	Bank lending standards: t +5.3 to 2011, 0.0 after.
H8c	Failed	Factory activity: no relation in either period.

Why the input-price claim failed

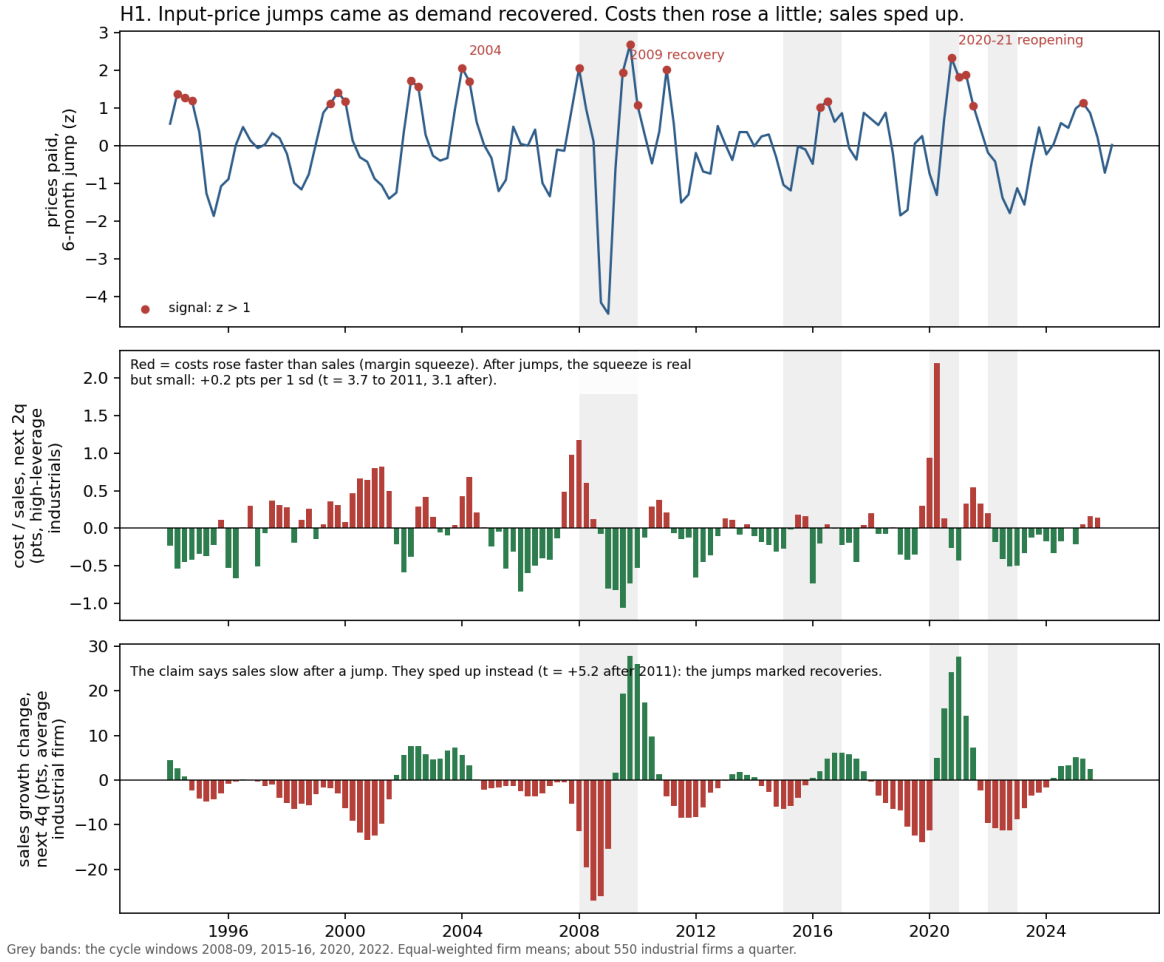


Figure 2: Top: the 6-month jump in prices paid; red dots are signals. Middle: the cost ratio of high-leverage industrial companies over the next 2 quarters. Bottom: the change in sales growth of the average industrial company over the next 4 quarters. The jumps came as demand recovered (2004, 2009, 2020–21). Sales then sped up.

The notes read a jump in input prices as a coming squeeze. In the data, most jumps came at the start of a recovery. Demand pushed input prices and sales up together. The squeeze on costs was real, but small, and it showed mainly in high-leverage industrial companies.

Do company sensitivities persist? (test C)

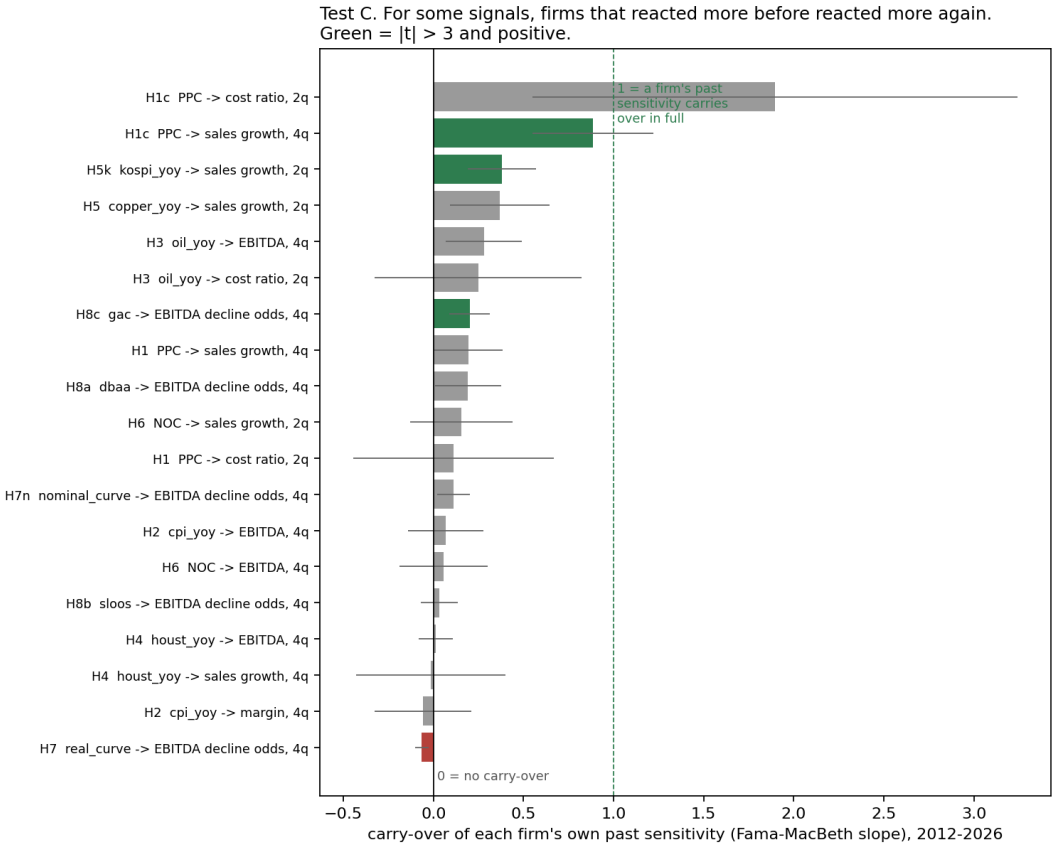


Figure 3: For each signal and outcome, the carry-over of each company's own past sensitivity, after 2011. Bars show ±2 standard errors.

For input-price jumps and sales (slope 0.89, t 5.3), the Korean stock market and sales (0.38, t 4.0), and factory activity and EBITDA declines (0.20, t 3.6), companies that reacted more before reacted more again. For most other signals the carry-over is near zero.

4 What it means for the model

1. **Add no Wolfe Trahan series to the forecast inputs m_t of the downside model.** None improves the forecast of company results reliably. The two passes are small (H1c) or borderline (H5k).
2. **Use the confirmed directions in the stress scenarios.** In an inflation path, lower consumer EBITDA and margins (H2). In an oil path, raise energy EBITDA and consumer costs (H3). These are signs from history, not forecasts.
3. **Review the model's own macro inputs.** After 2011, bank lending standards and the factory survey did not predict company EBITDA declines. The Baa spread did a little. The downside model (methodology 5.2) uses all three. Open a decision page and rerun this test on the issuer universe before the model relies on them.
4. **Estimate sensitivity per issuer where it persists.** For input prices and global demand, past company sensitivity carries over. A sensitivity per issuer, shrunk toward its sector, may beat one sensitivity per sector (the current `fit_sector_beta`). Test that before use.

5 Limits

1. The company data is Compustat's current view, after restatements. It shows what happened, not what investors saw first.
2. Sales are in dollars, so inflation raises them. Cost ratios and margins do not have this bias.
3. Some groups are small: about 55 housing-linked and 40 auto companies a quarter. The high-leverage tests for these groups have too few companies after 2011.
4. With 87 tests, about 4 would show $p < 0.05$ by chance. Only 1 survives the strictest correction (Holm across all 87 tests).
5. GICS codes are today's. A company that changed sector counts in its current one.
6. The signal transforms follow the charts in the notes. H1c is a second transform of the same claim.

Reproduce

Results go to `data/raw/backtest/wt_firm_*.json`.

```
PYTHONPATH=. python3 scripts/wt_tests.py     # all claims
PYTHONPATH=. python3 scripts/wt_figures.py   # the figures
python3 -m pytest -q tests/test_wt_tests.py  # 12 tests
```

Inputs: `data/raw/compustat/fundq_nonfin.parquet` (WRDS, `comp.fundq`), `data/raw/macro/` (Philadelphia Fed survey, FRED). Vendor data stays out of git.